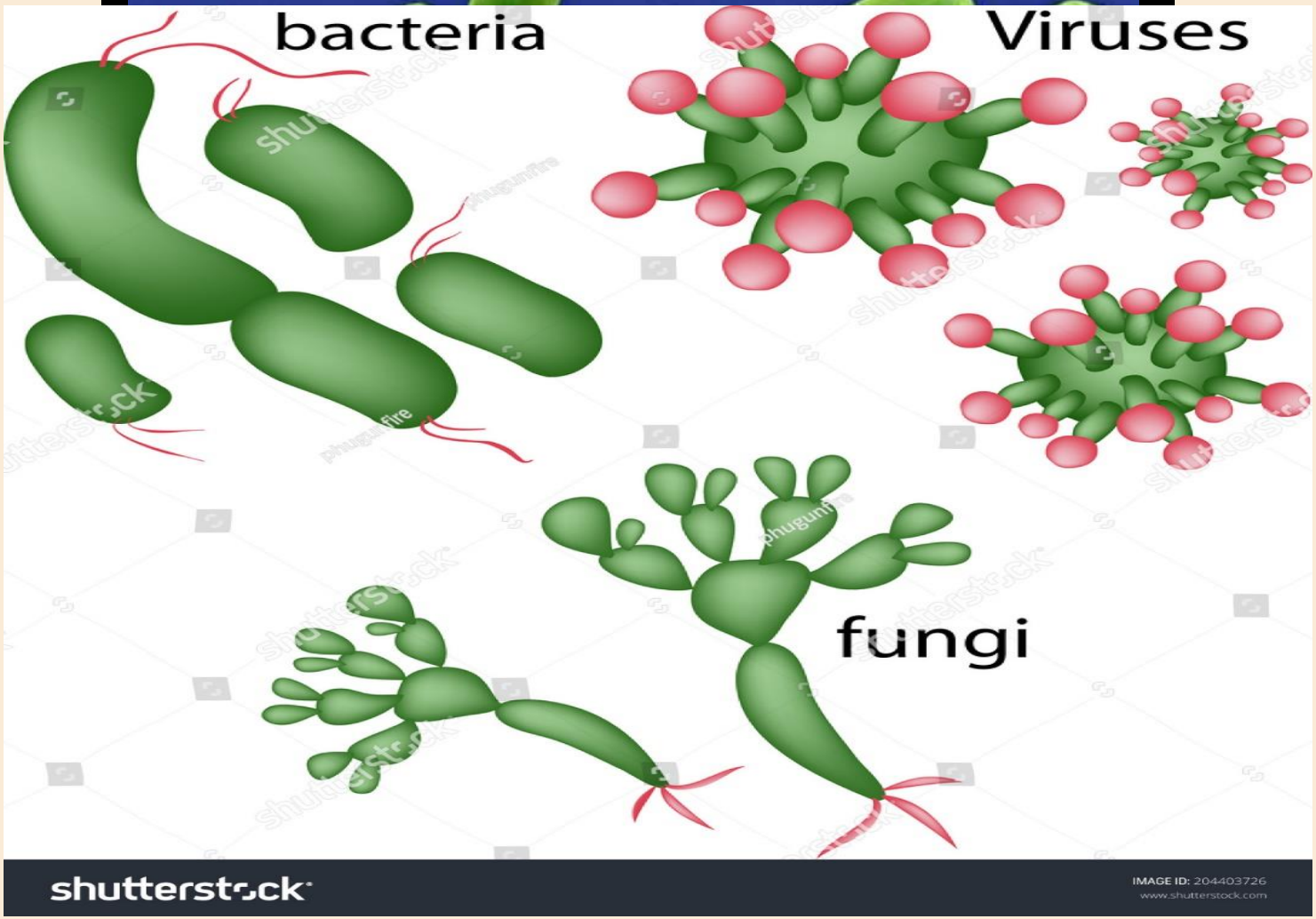
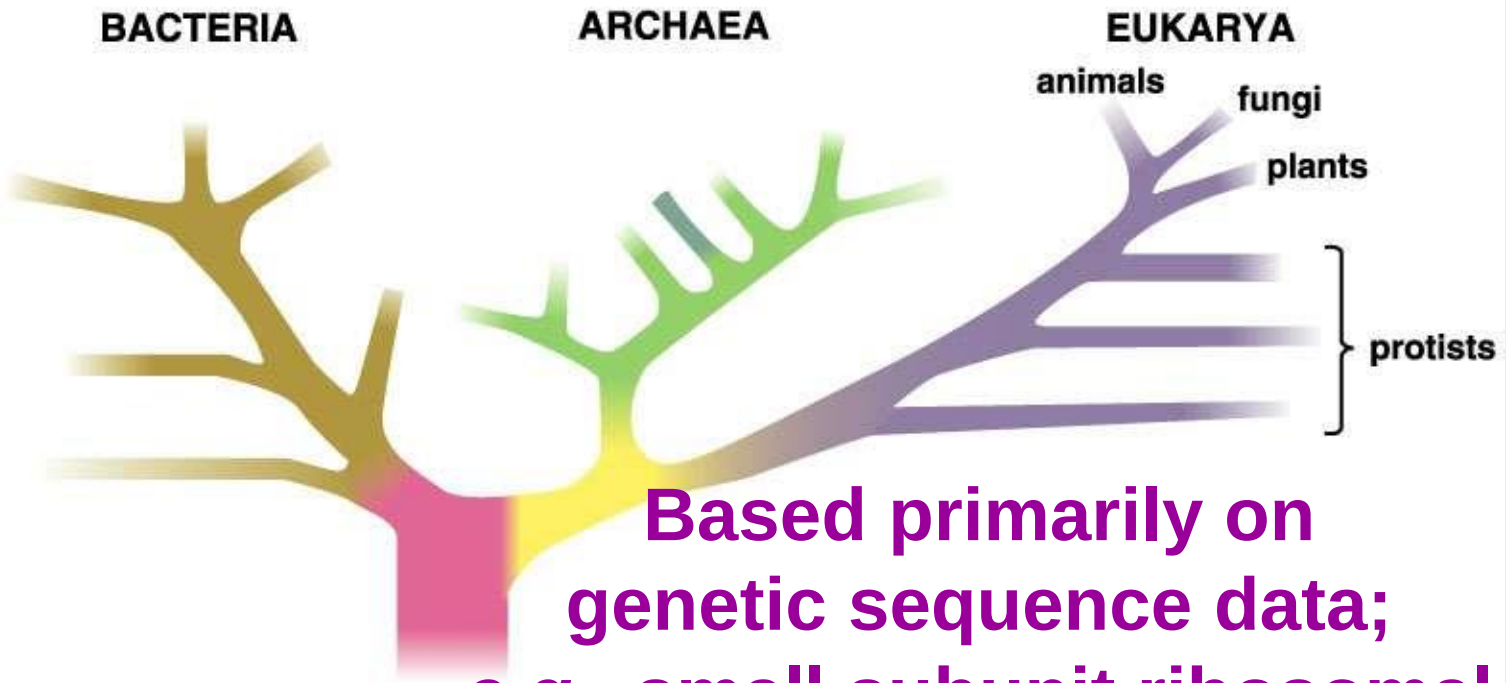
Lec 1: Microbial Diversity

Prof. Dr. Marwa Salah
Professor of microbiology
Microbial Biotechnology Department
Faculty of Biotechnology –University of Sadat city

Microbial diversity



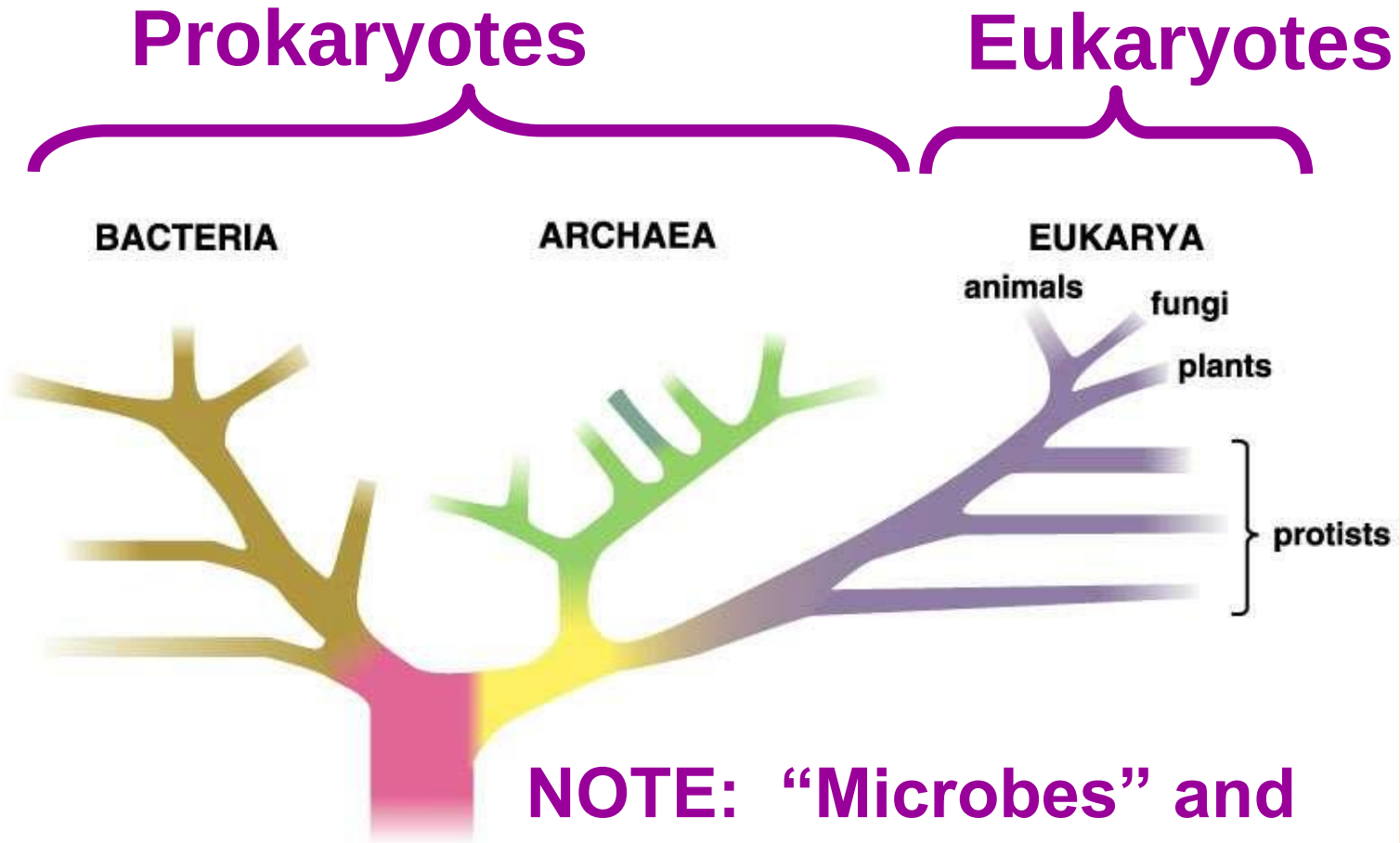
Carl Woese's 3 Domains of Life



Based primarily on genetic sequence data; *e.g.*, small subunit ribosomal RNA – present in all organisms

See Fig. 27.12

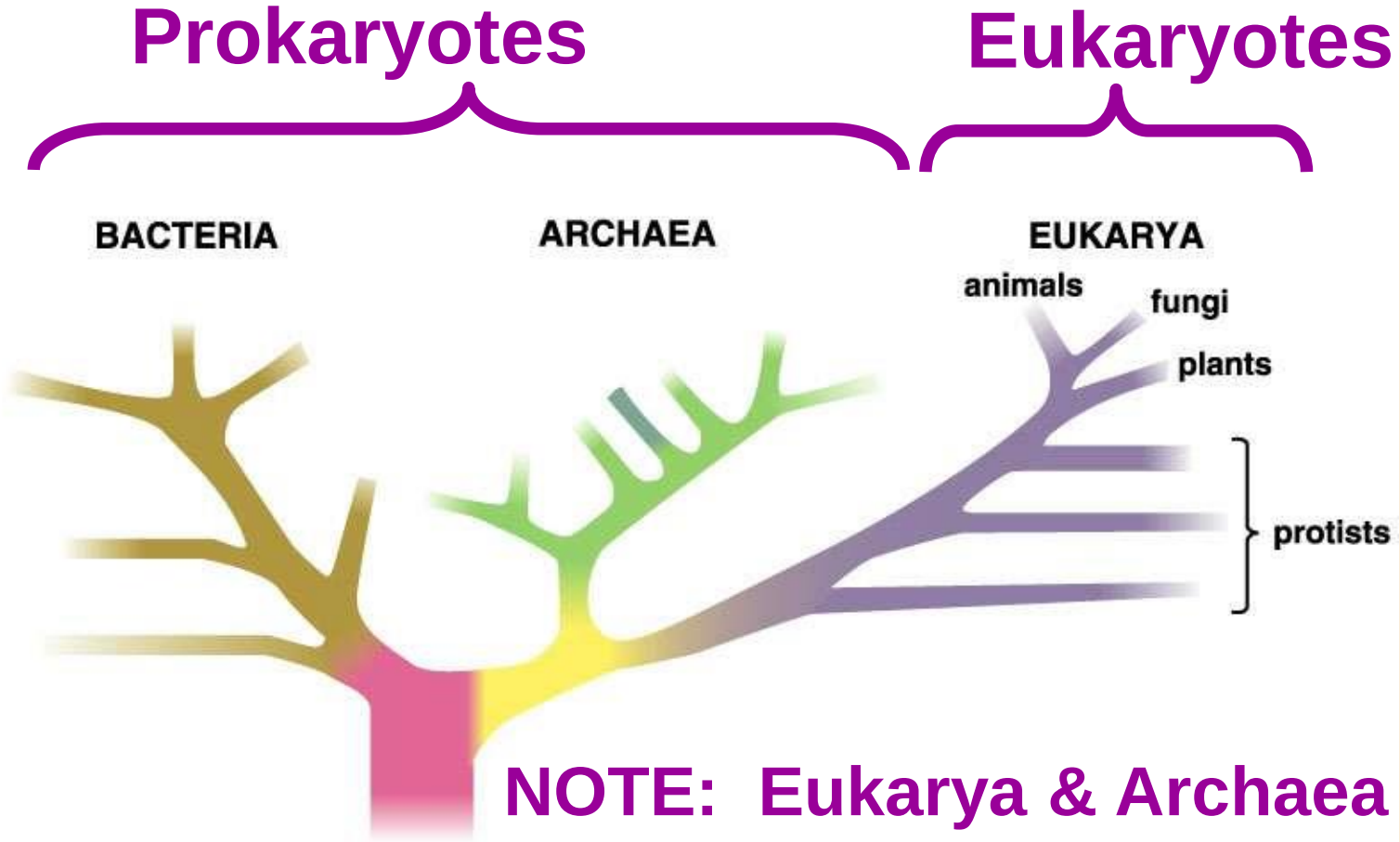
Carl Woese's 3 Domains of Life



NOTE: "Microbes" and "Prokaryotes" are not taxonomic categories

See Fig. 27.12

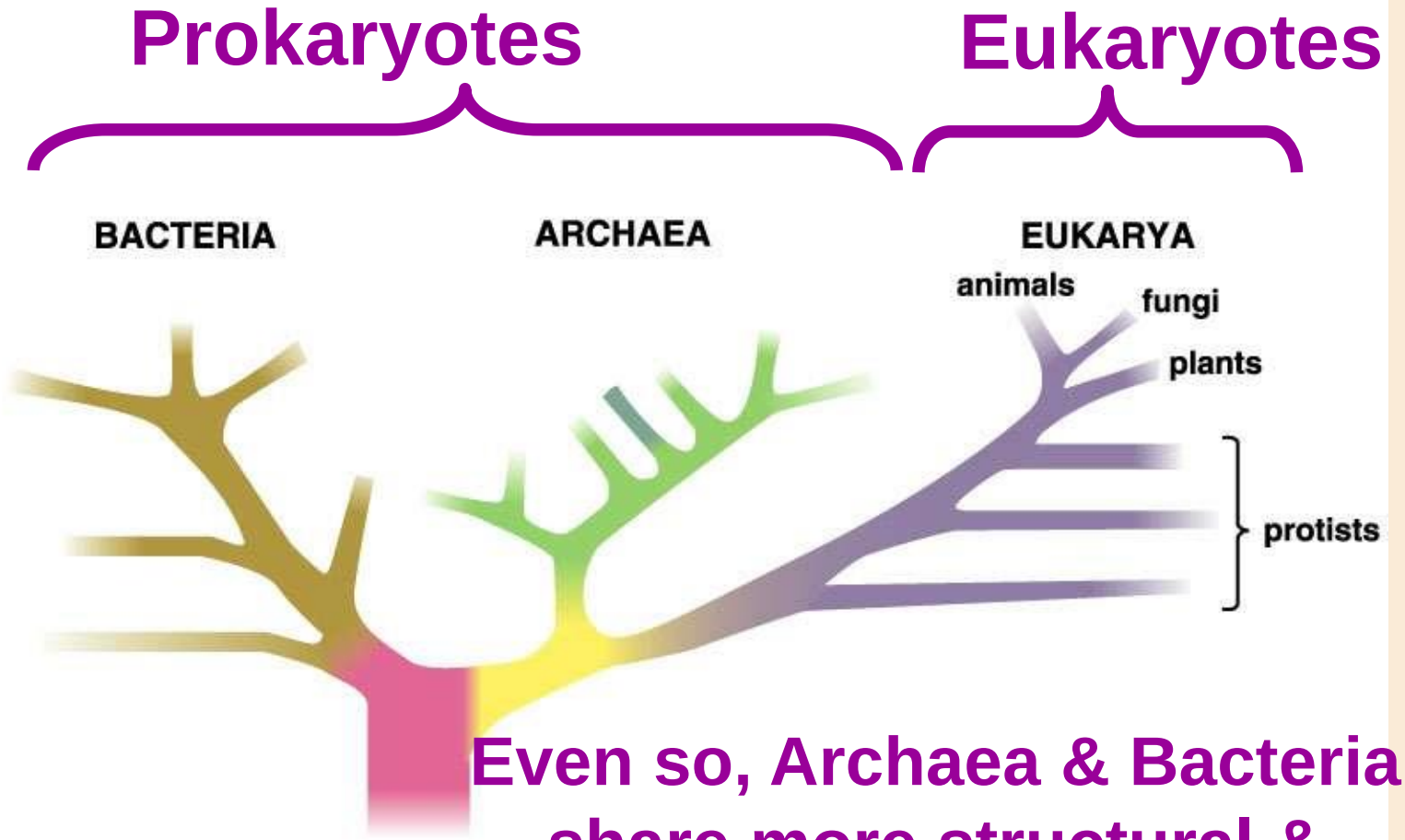
Carl Woese's 3 Domains of Life



NOTE: Eukarya & Archaea are more closely related than either is to Bacteria

See Fig. 27.12

Carl Woese's 3 Domains of Life



Even so, Archaea & Bacteria share more structural & functional properties than either does with Eukarya

See Fig. 27.12

What are microbes?

Single-celled organisms and some non-cellular parasites

Kinds of microbes

Non-cellular, parasitic molecules

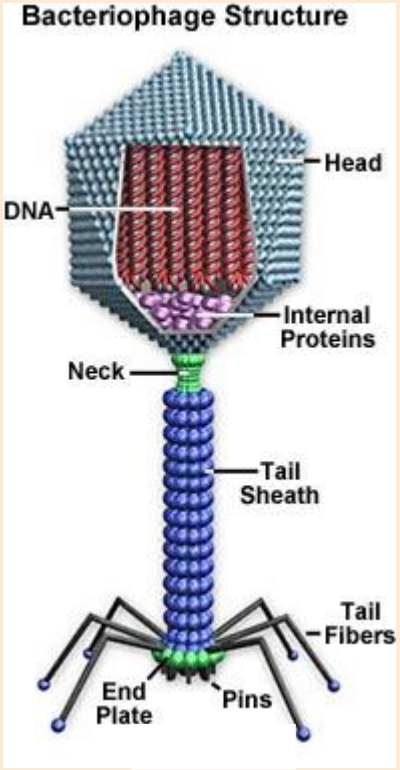
- Viruses
- Viroids
- Prions

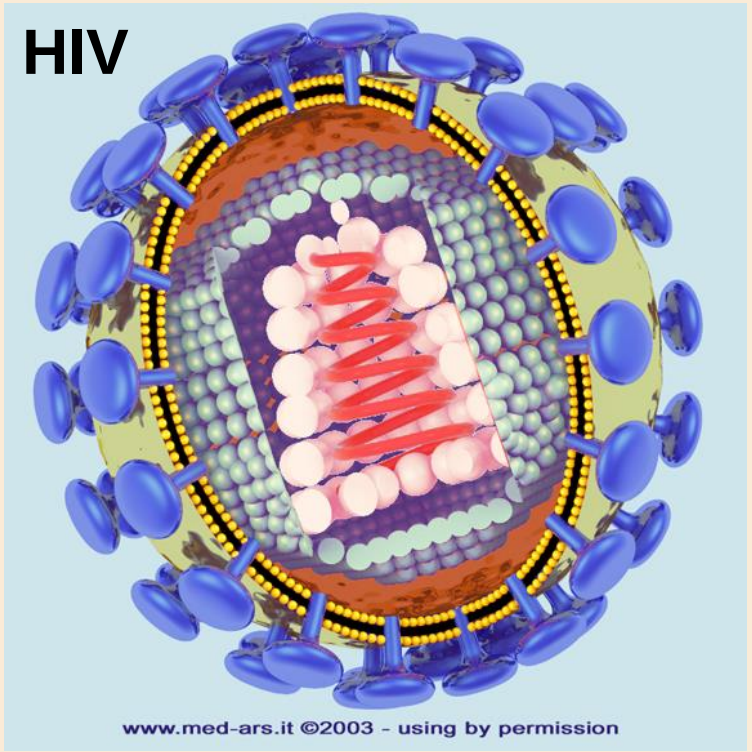
Nonliving parasitic molecules

Viruses

Single or double stranded RNA or DNA with a protein coat

Common cold, Ebola, HIV





Nonliving parasitic molecules

Viruses

Single or double stranded RNA or DNA with a protein coat

Common cold, Ebola, HIV

Viroids

Short, single strand of RNA w/o protein coat

Primarily infect plants

Prions

Protein particles w/o genetic material

Kuru, mad cow, chronic wasting disease

Kinds of microbes

Non-cellular, parasitic molecules

- Viruses
- Viroids
- Prions

Prokaryotes

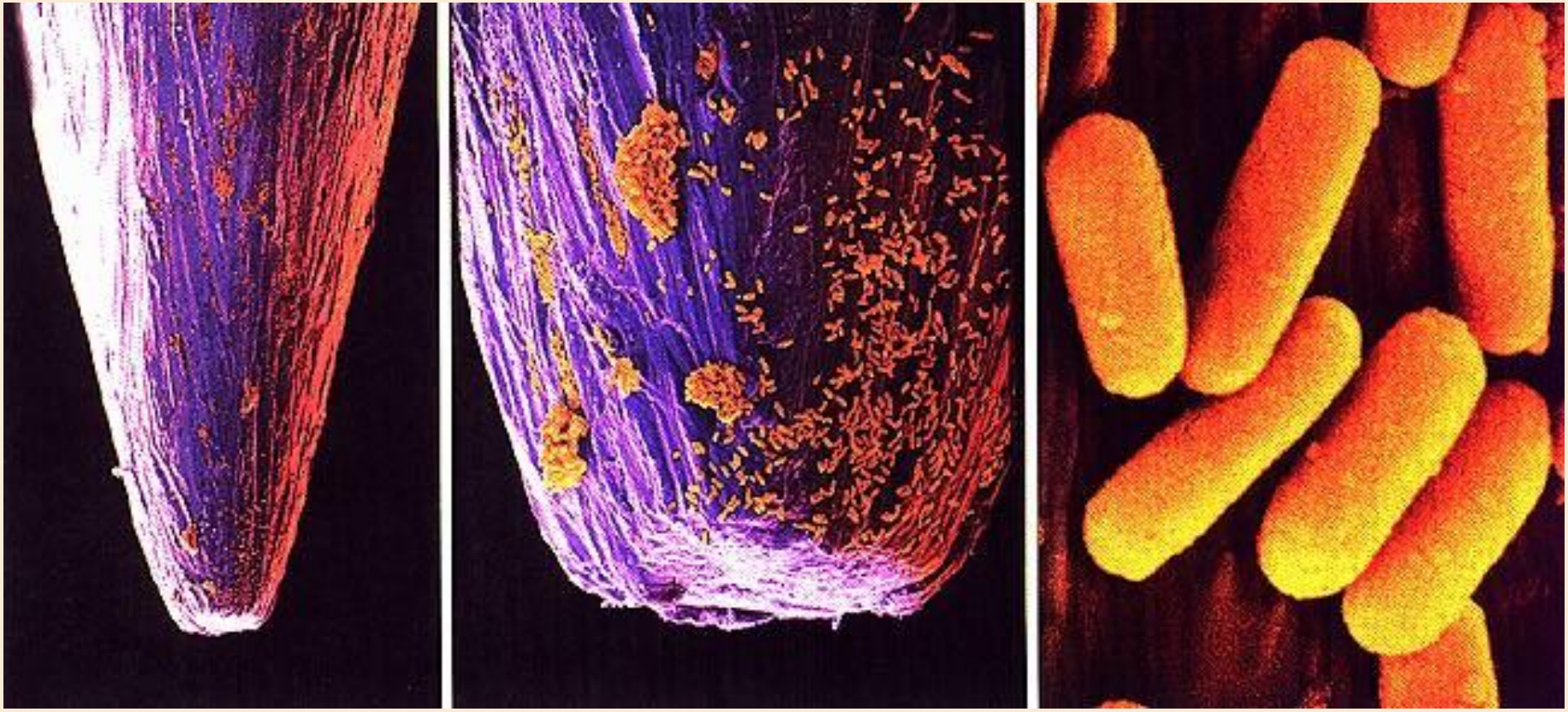
- Domain Bacteria
- Domain Archaea

Eukaryotes

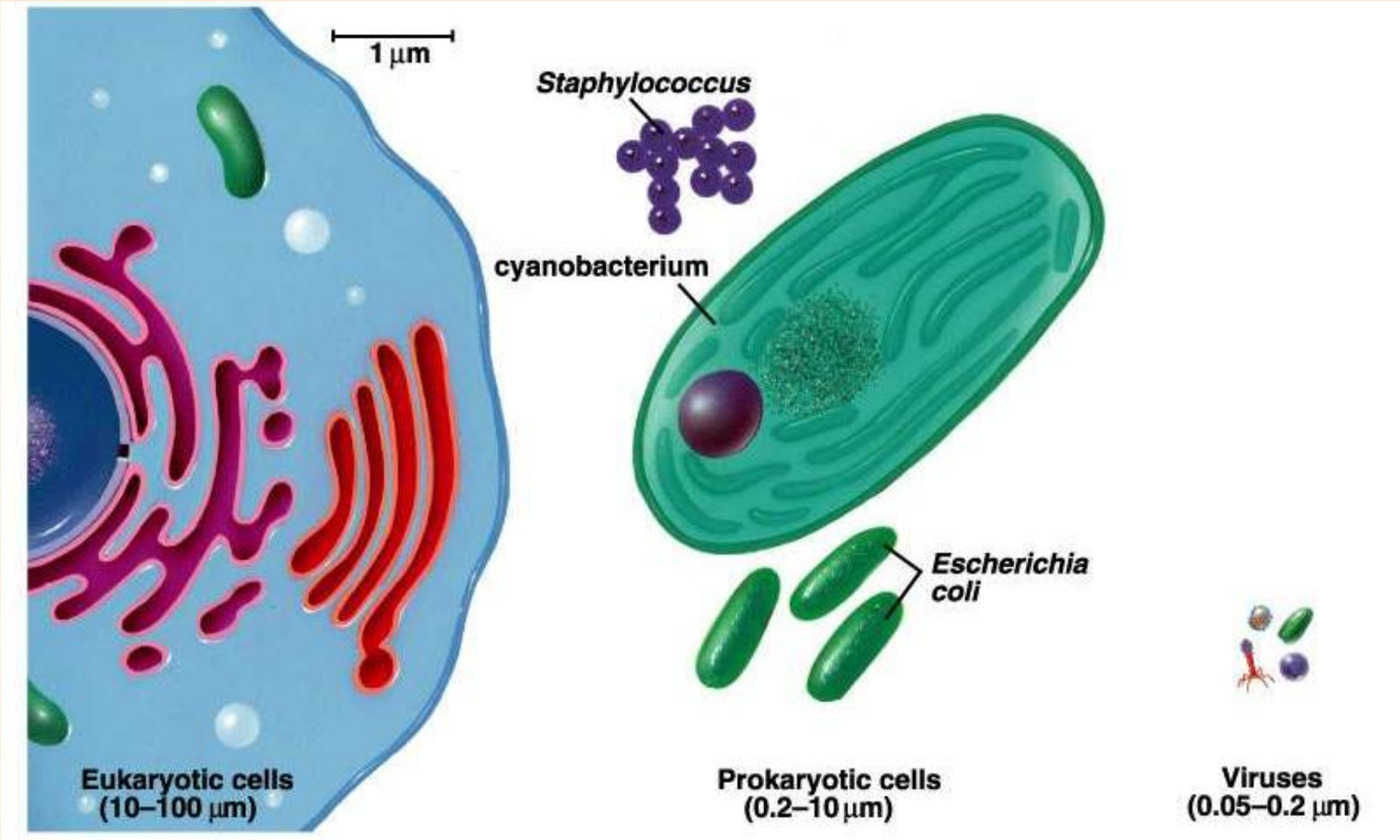
- Several Kingdoms in Domain Eukarya

Microbes are microscopic, but very diverse & extremely abundant

Bacteria on the tip of a pin



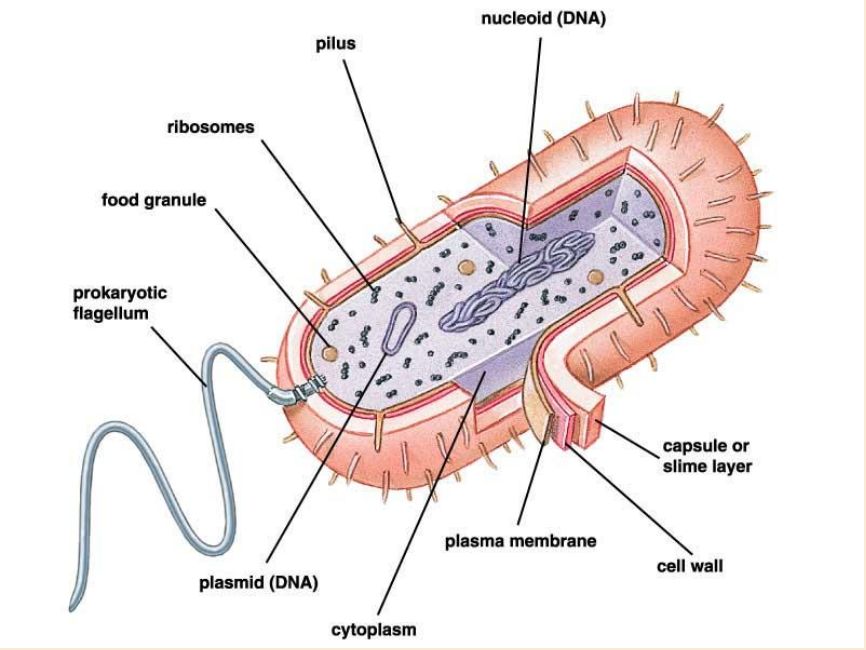
Relative sizes of microbes



Prokaryotes

Domains Bacteria & Archaea

Simple cells – with no nucleus or membrane-bound organelles



Prokaryotes

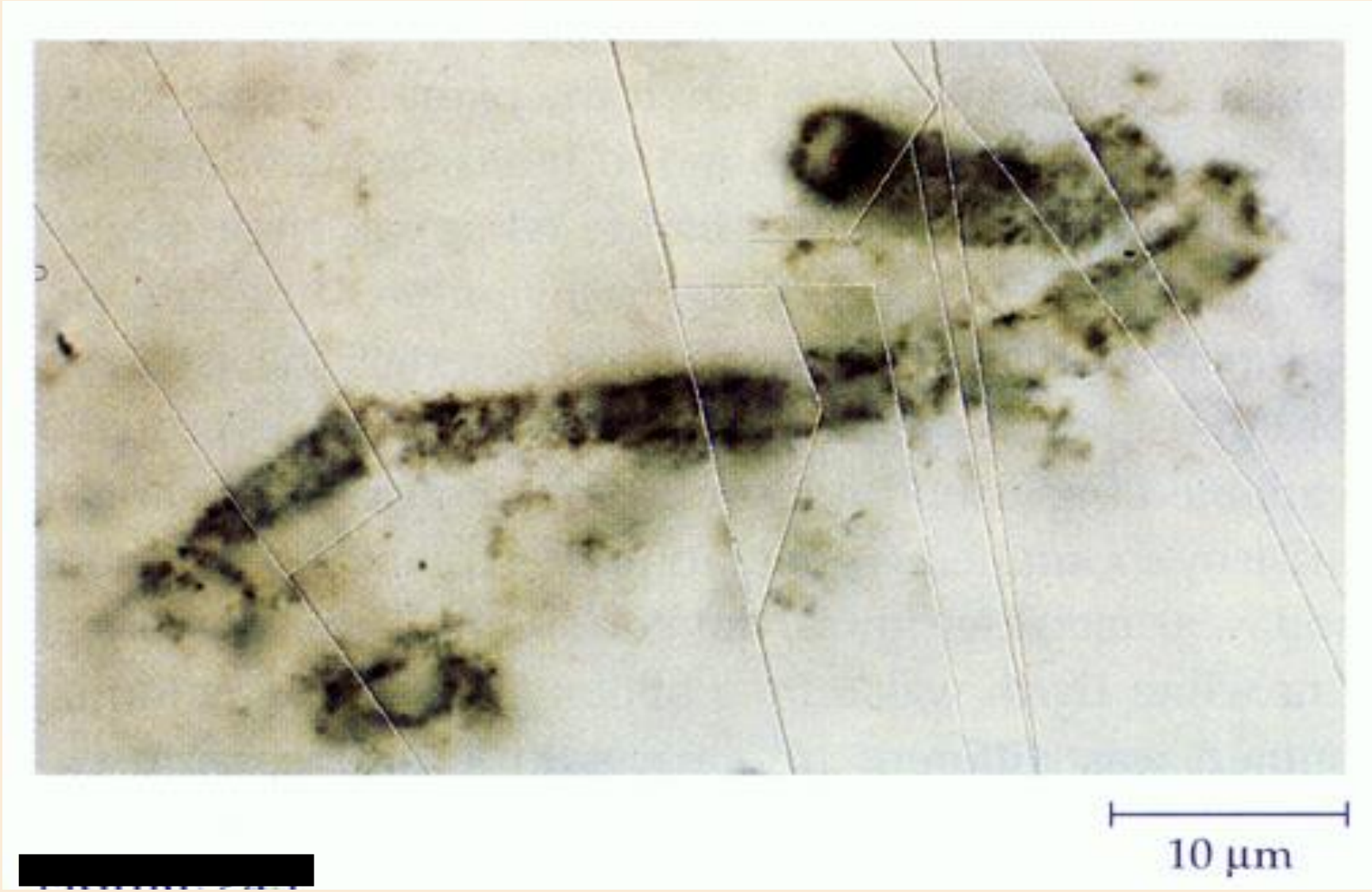
Domains Bacteria & Archaea

Simple cells – with no nucleus or membrane-bound organelles

First organisms on Earth – at least 3 billion years ago

The oldest known fossils

Cyanobacteria > 3 billion years old



Prokaryotes

Domains Bacteria & Archaea

- Simple cells – with no nucleus or membrane-bound organelles
- First organisms on Earth – at least 3 billion years ago
- **Distributed globally – including many extremophiles**

Extreme Thermophiles

"Heat-loving" *Archaea*



Extreme Halophiles

"Salt-loving" *Archaea*



Australian Salt Lake

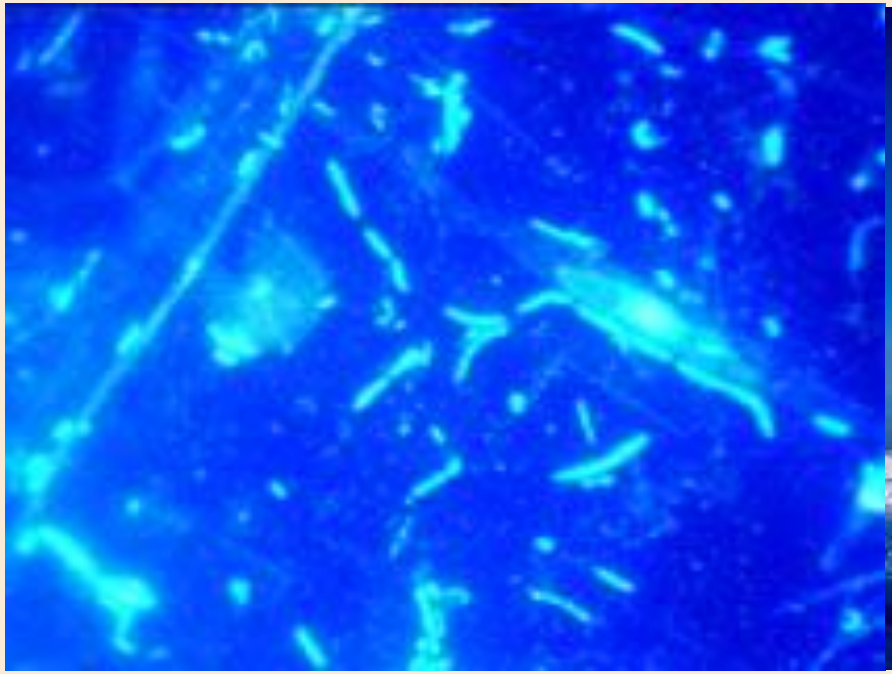
Methanogens
Methane-generating *Archaea*

Occur in oxygen-free habitats

***E.g.*, swamp mud, guts of ruminant animals**



Ice Bacteria & Archaea





Cave Bacteria

Sometimes reaching acidity of pH 0.5



Prokaryotes

Domains Bacteria & Archaea

- Simple cells – with no nucleus or organelles
- First organisms on Earth – at least 3 billion years ago
- Distributed globally – including many extremophiles
- **Nutrition – autotrophs & heterotrophs**

Prokaryote Nutrition

All organisms require a source of energy & carbon

Autotrophs can obtain all their C from CO_2

Table 27.1 Major Nutritional Modes

Mode of Nutrition	Energy Source	Carbon Source	Types of Organisms
Autotroph			
Photo-autotroph	Light	CO_2	Photosynthetic prokaryotes, including cyanobacteria; plants; certain protists (algae)
Chemo-autotroph	Inorganic chemicals	CO_2	Certain prokaryotes (for example, *Sulfolobus*)
Heterotroph			
Photo-heterotroph	Light	Organic com-pounds	Certain prokaryotes
Chemo-heterotroph	Organic com-pounds	Organic com-pounds	Many prokaryotes and protists; fungi; animals; some parasitic plants

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Prokaryote Nutrition

All organisms require a source of energy & carbon

Heterotrophs require at least one organic nutrient, *e.g.*, glucose

Table 27.1 Major Nutritional Modes

Mode of Nutrition	Energy Source	Carbon Source	Types of Organisms
Autotroph			
Photoautotroph	Light	CO_2	Photosynthetic prokaryotes, including cyanobacteria; plants; certain protists (algae)
Chemoautotroph	Inorganic chemicals	CO_2	Certain prokaryotes (for example, *Sulfolobus*)
Heterotroph			
Photoheterotroph	Light	Organic compounds	Certain prokaryotes
Chemoheterotroph	Organic compounds	Organic compounds	Many prokaryotes and protists; fungi; animals; some parasitic plants

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Prokaryote Nutrition

All organisms require a source of energy & carbon

Phototrophs obtain their energy from the sun

Table 27.1 Major Nutritional Modes

Mode of Nutrition	Energy Source	Carbon Source	Types of Organisms
Autotroph			
Photoautotroph	Light	CO_2	Photosynthetic prokaryotes, including cyanobacteria; plants; certain protists (algae)
Chemoautotroph	Inorganic chemicals	CO_2	Certain prokaryotes (for example, *Sulfolobus*)
Heterotroph			
Photoheterotroph	Light	Organic compounds	Certain prokaryotes
Chemoheterotroph	Organic compounds	Organic compounds	Many prokaryotes and protists; fungi; animals; some parasitic plants

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Prokaryote Nutrition

All organisms require a source of energy & carbon

Chemotrophs obtain their energy from chemical compounds

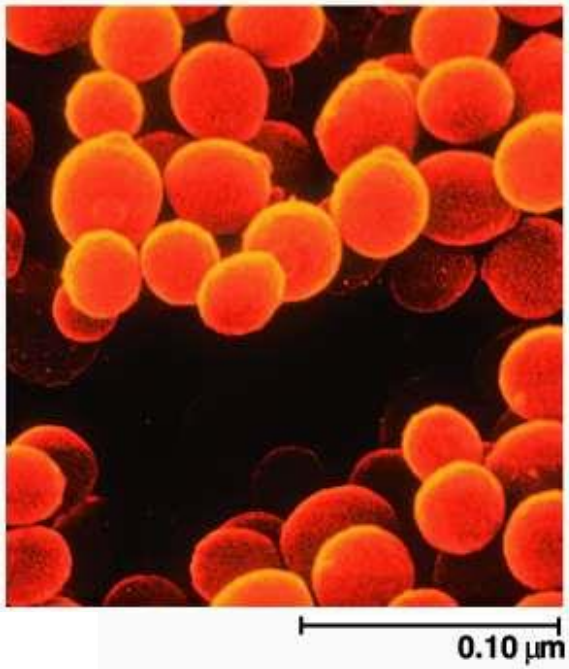
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Photoheterotroph	Light	Organic compounds	Certain prokaryotes
Chemoheterotroph	Organic compounds	Organic compounds	Many prokaryotes and protists; fungi; animals; some parasitic plants

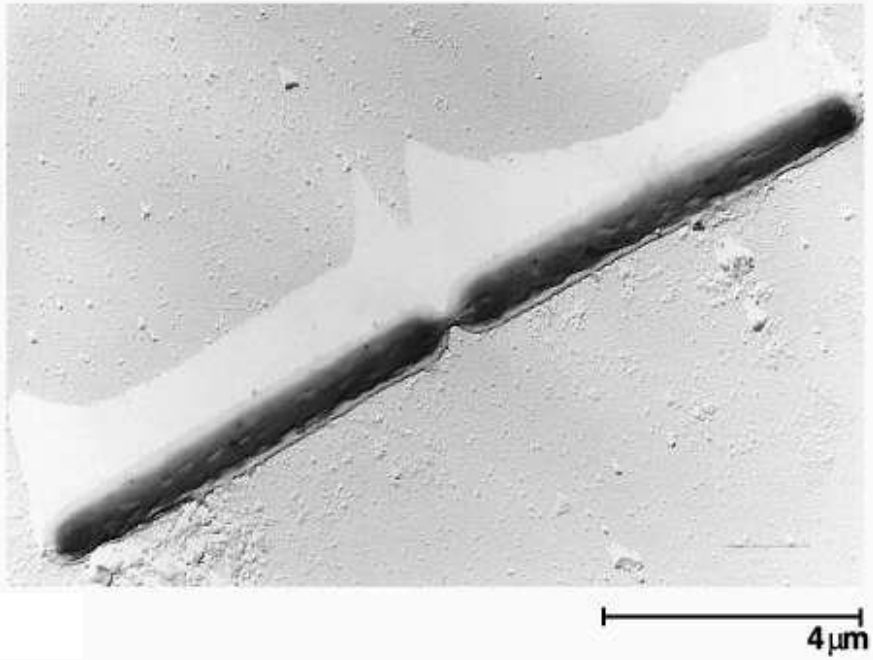
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3 common bacterial shapes:

Cocci- spheres



Bacilli- rods



Spirilli- spirals

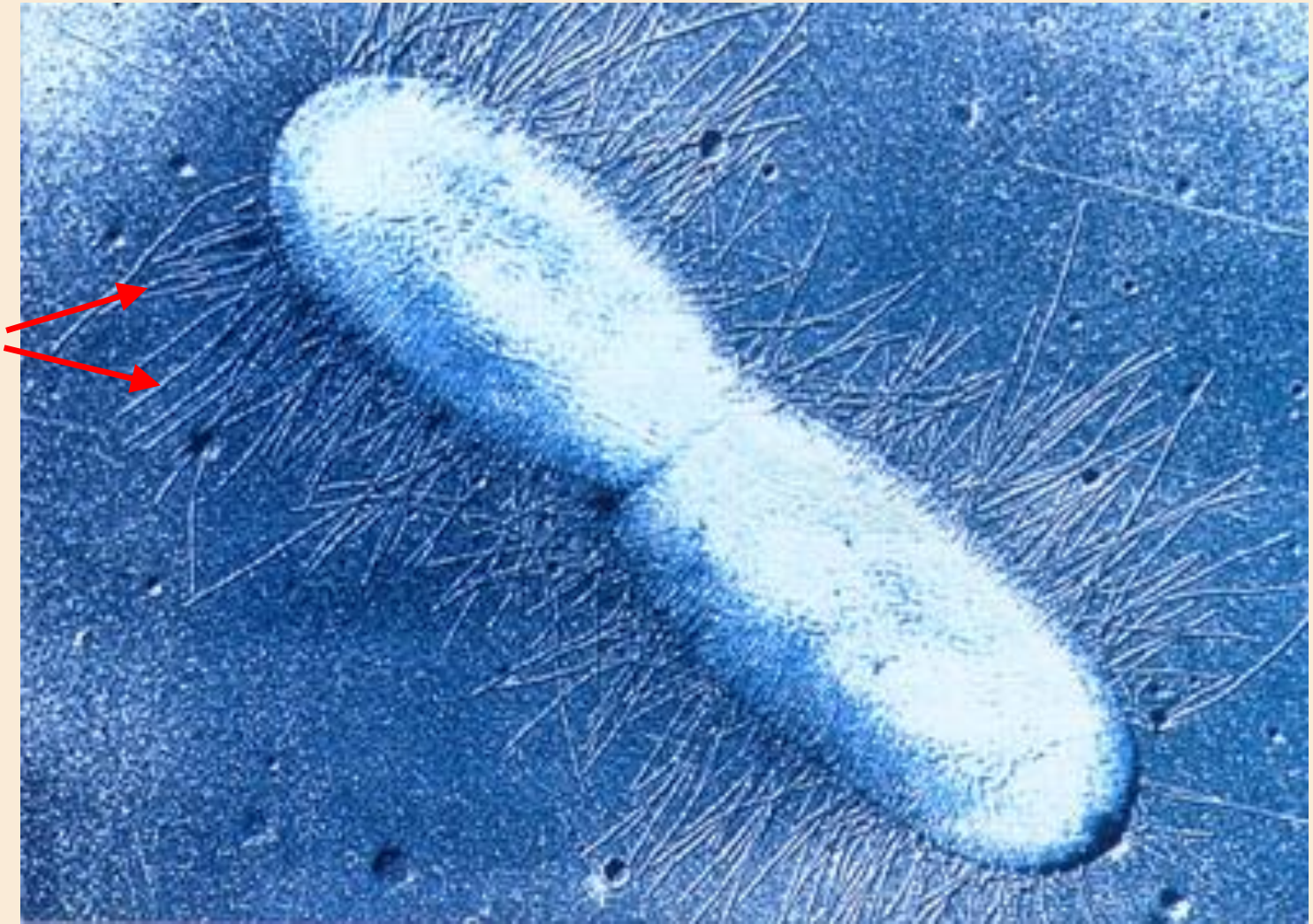


E.g., slime layer allows bacteria to cling to tooth enamel or other substrates



Pili (singular: pilus)

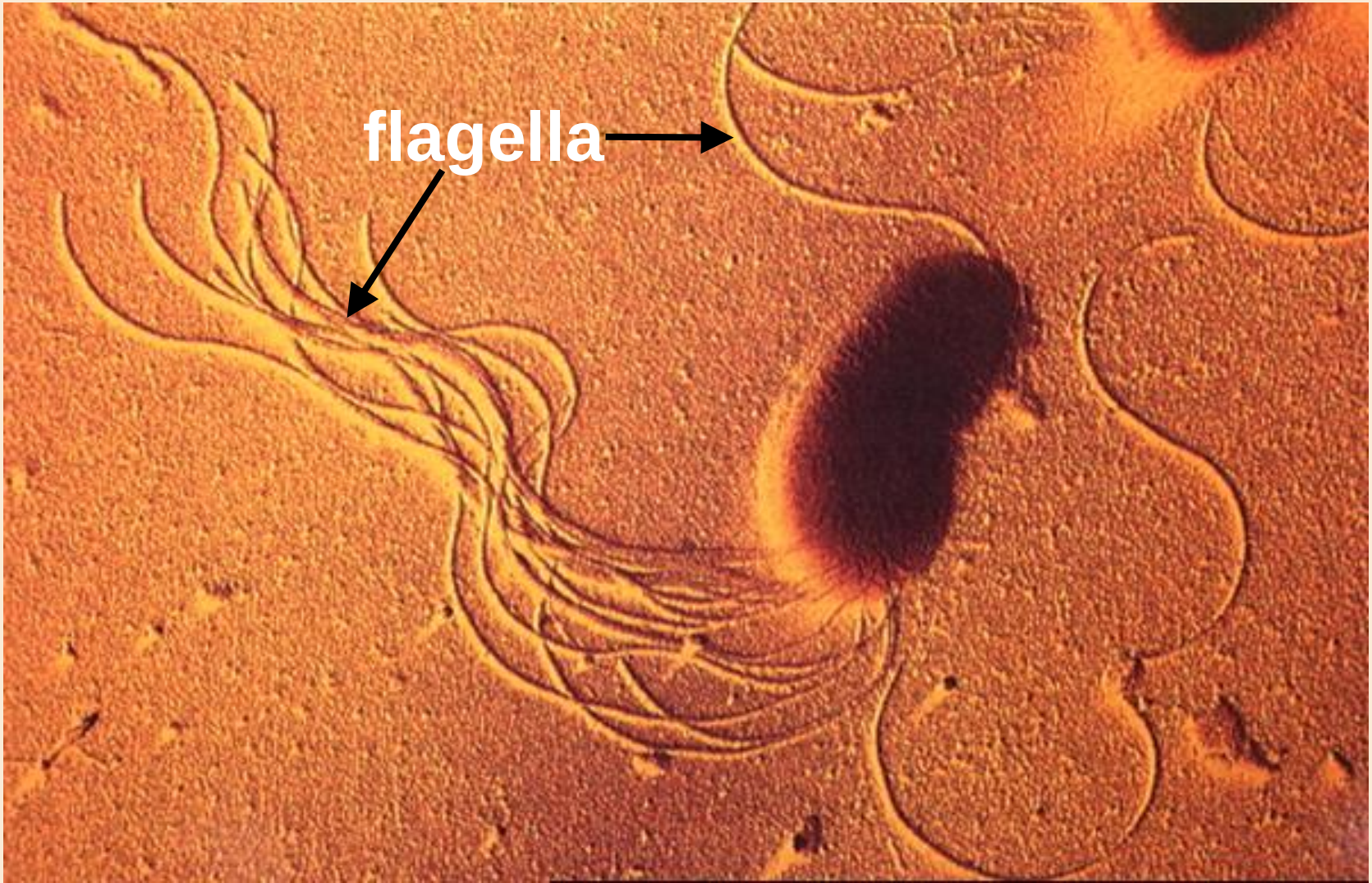
Protein filaments that attach bacteria to other cells & substrates



Some prokaryotes have flagella

(singular: flagellum)

Used for locomotion



Base of a bacterial flagellum…

…the only known wheel in nature

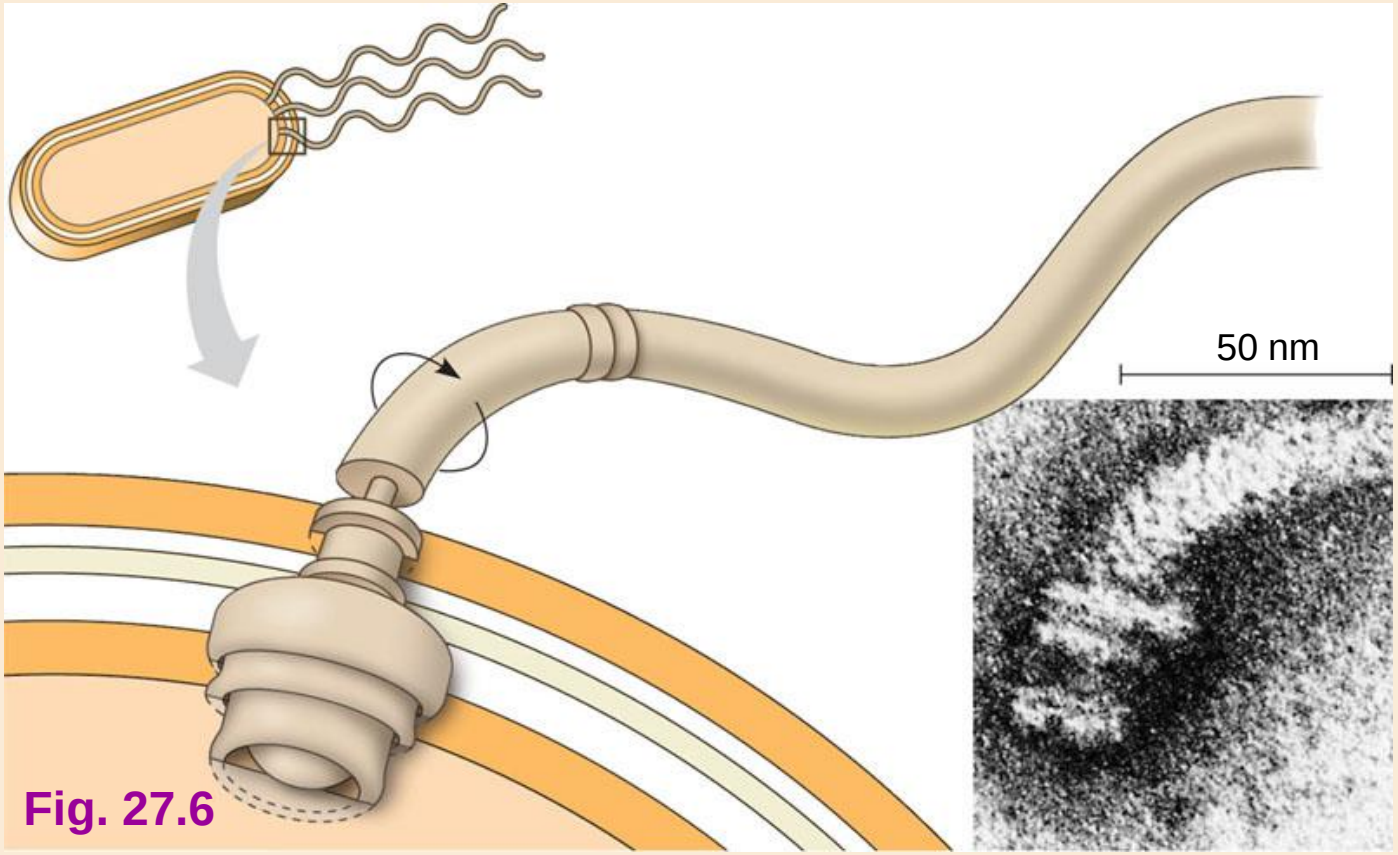


Fig. 27.6

What is “taxis”?

Motility allows some bacteria to move towards or away from stimuli

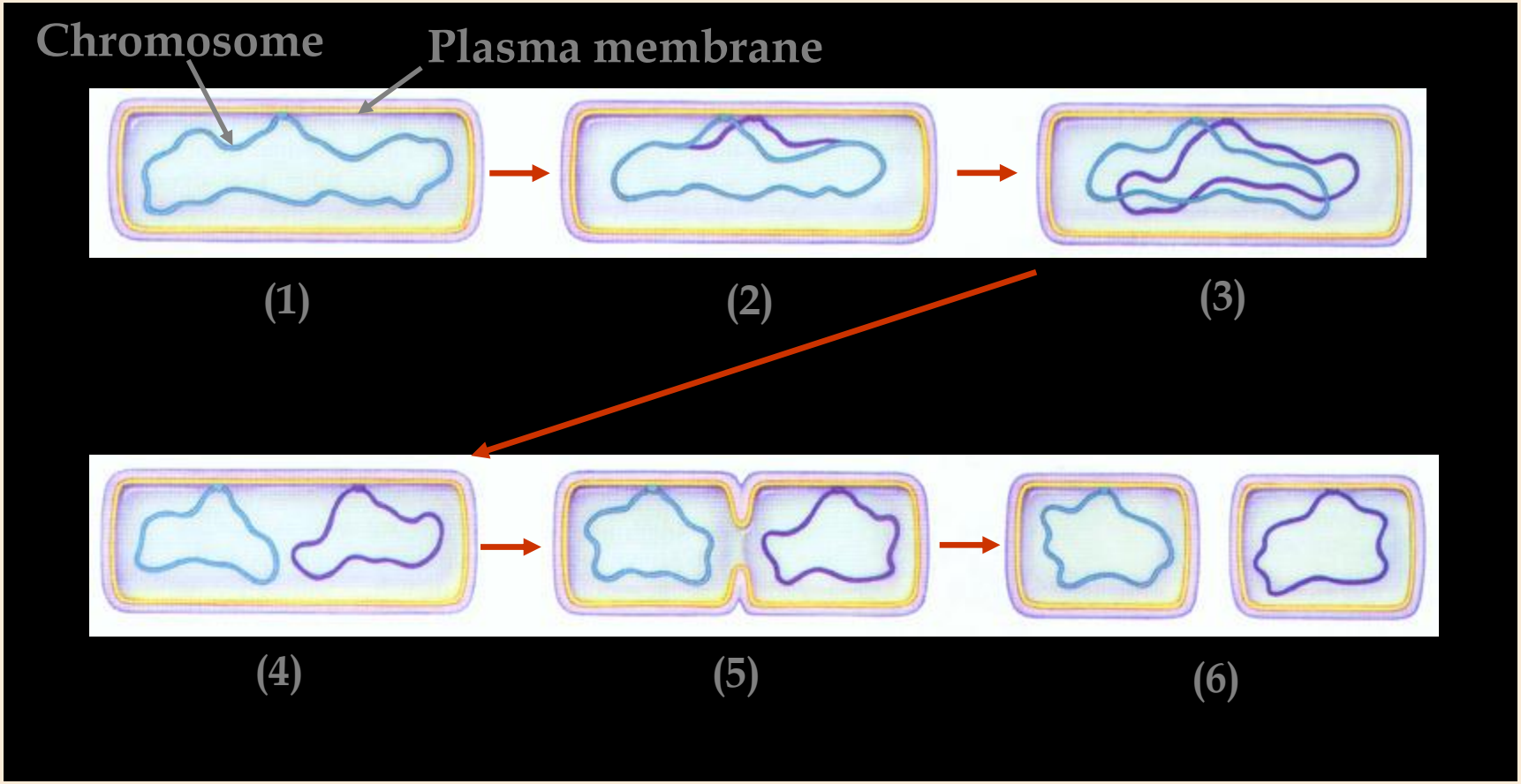
Phototaxis

Chemotaxis

Magnetotaxis

Binary fission

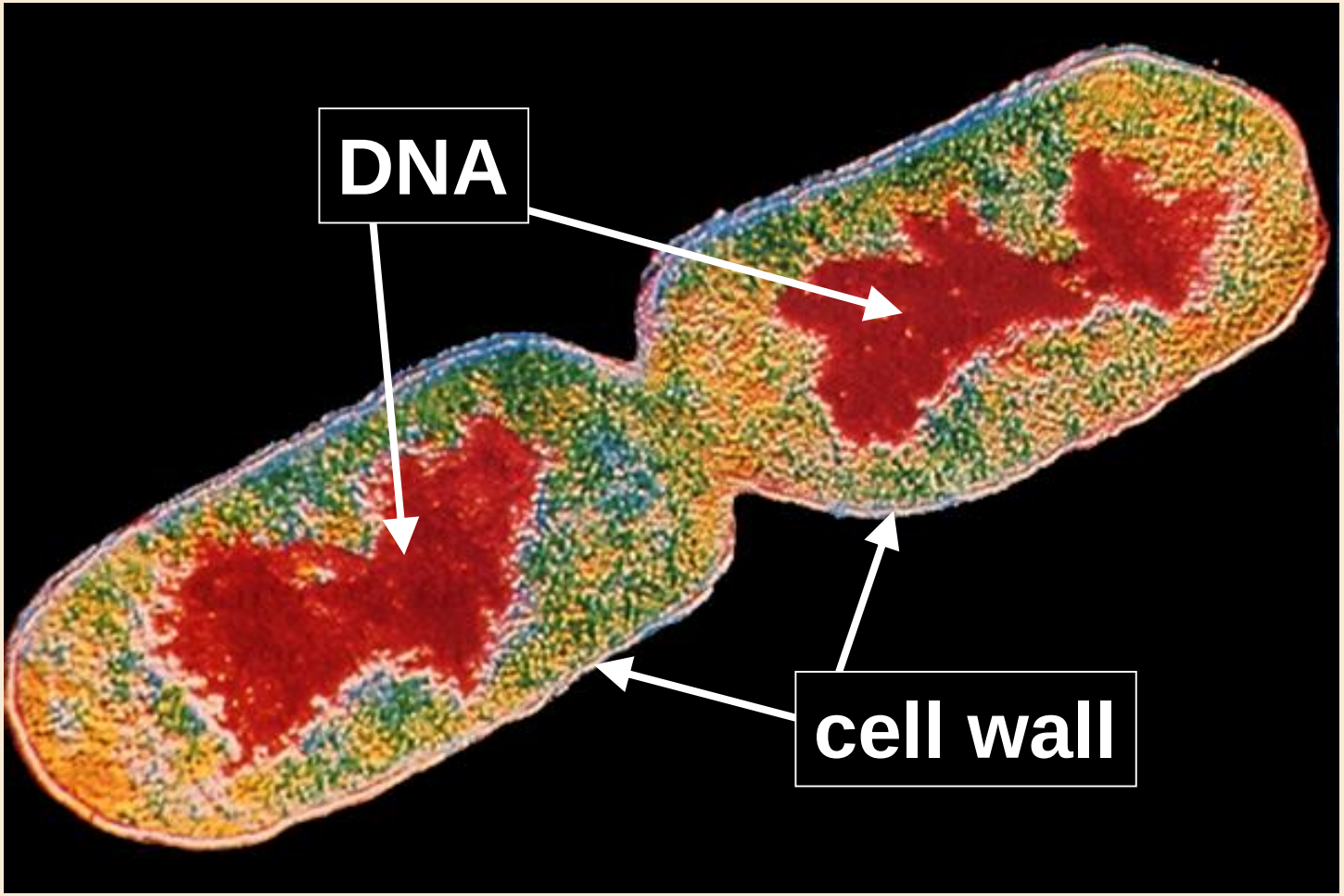
Daughter cells are identical copies



Neither mitosis nor meiosis occurs in prokaryotes

Binary fission

E. coli



Bacteria

Reproduction

Asexual, through binary fission

No true sexual reproduction, since neither mitosis nor meiosis exist in prokaryotes

Bacteria

Reproduction

Asexual, through binary fission

No true sexual reproduction, since neither mitosis nor meiosis exist in prokaryotes

Horizontal transfer of genetic material

Transformation — **Uptake of genetic material from the environment**

Bacteria

Reproduction

Asexual, through binary fission

No true sexual reproduction, since neither mitosis nor meiosis exist in prokaryotes

Horizontal transfer of genetic material

Transformation

Transduction

Transfer of genetic material between prokaryotes by viruses

Bacteria

Reproduction

Asexual, through binary fission

No true sexual reproduction, since neither mitosis nor meiosis exist in prokaryotes

Horizontal transfer of genetic material

Transformation

Transduction

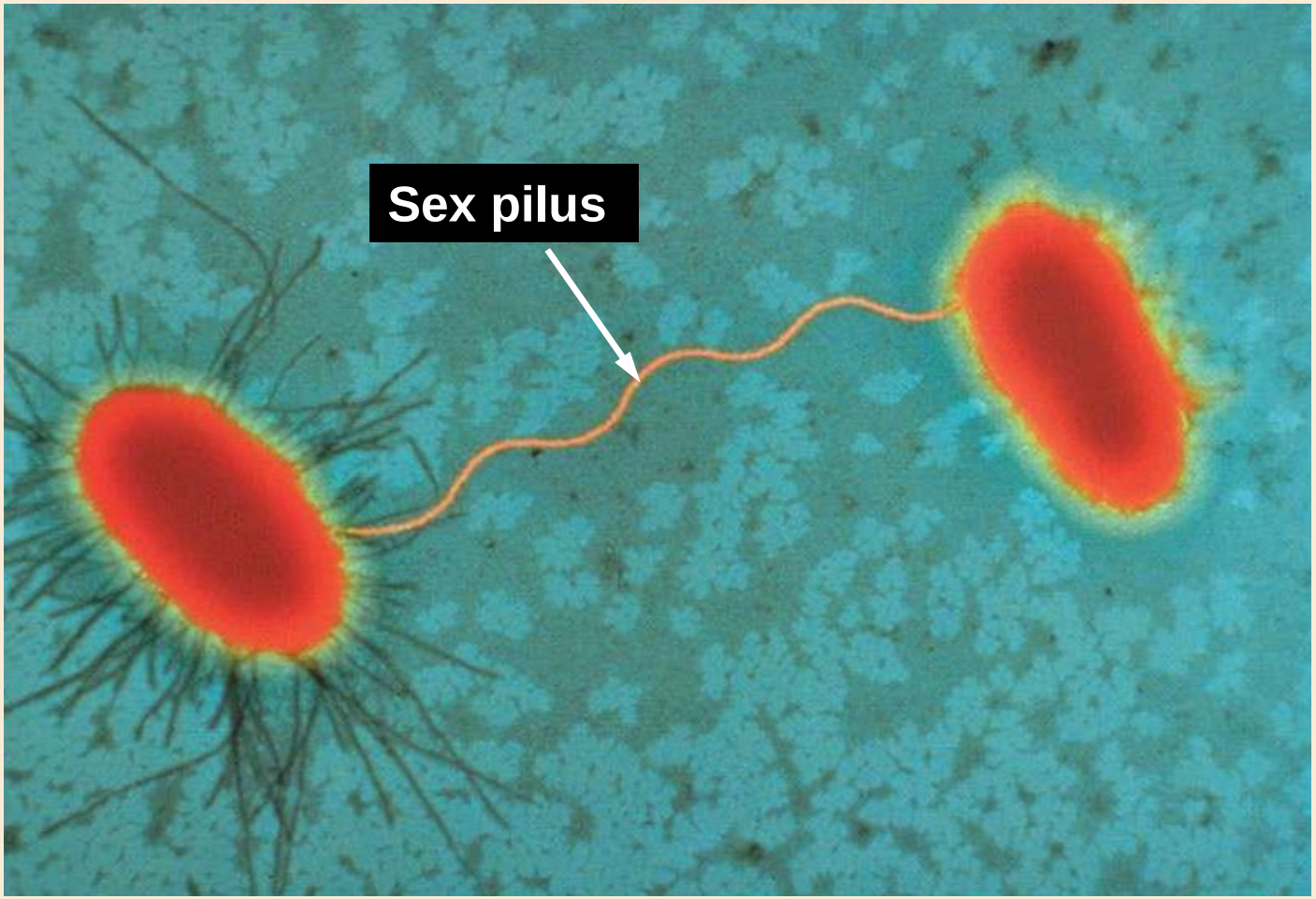
Conjugation

Direct transfer of genetic material from one prokaryote to another

Conjugation in *E. coli*

Sex pilus connects cells and draws them together

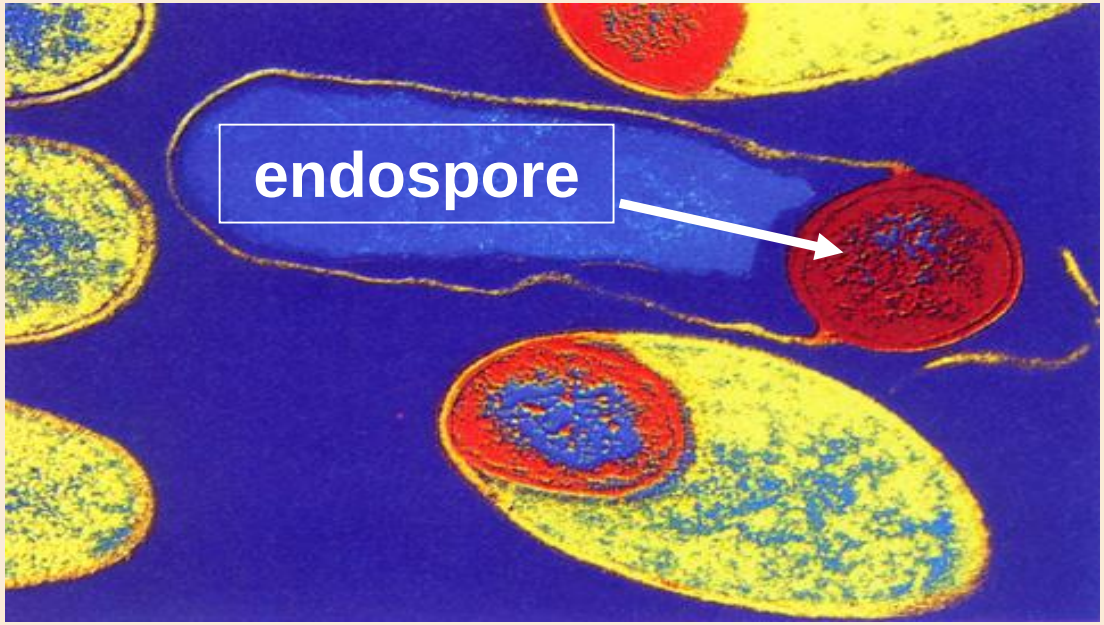
Conjugation tube then forms



Bacteria

Surviving harsh conditions

Endospore – forms inside a bacterium and then persists through inhospitable conditions



Bacteria – Impacts on other organisms, including Human Society

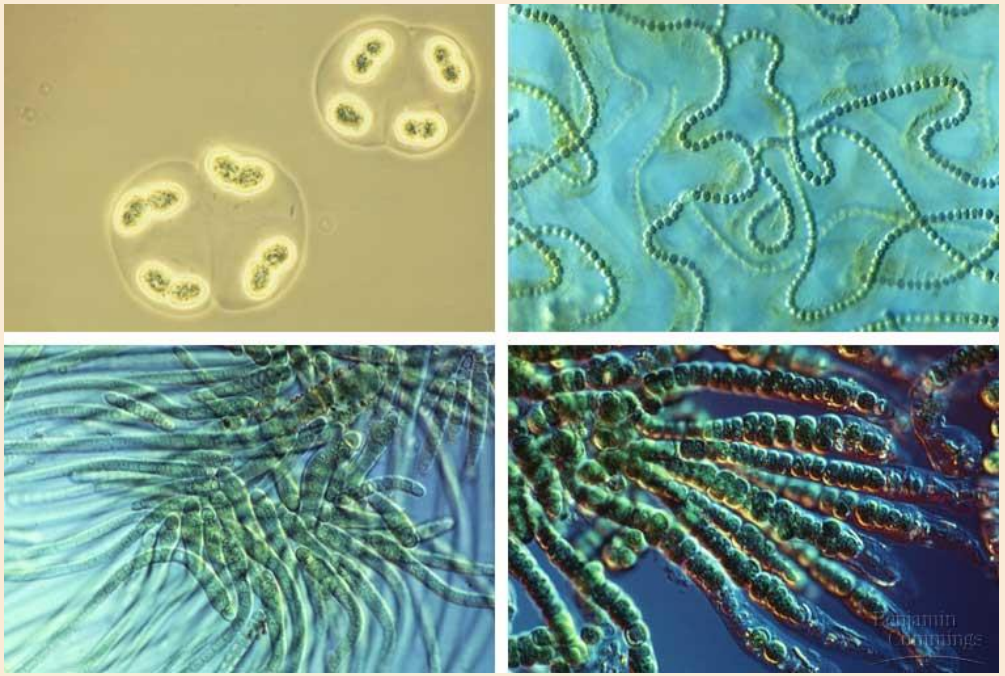
Decomposition

Bacteria – Impacts on other organisms, including Human Society

Decomposition

Photosynthesis

Especially common in the cyanobacteria



Bacteria – Impacts on other organisms, including Human Society

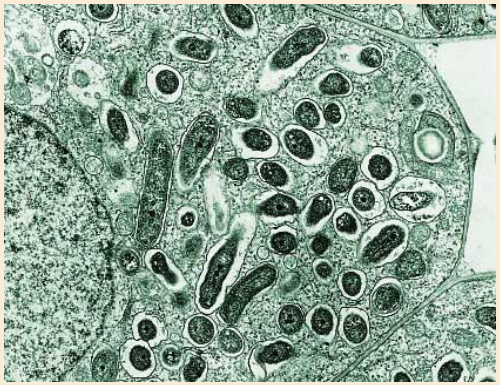
Decomposition

Photosynthesis

N-fixation

Symbiosis

Mutualism



Rhizobium **in the roots of a soybean plant**

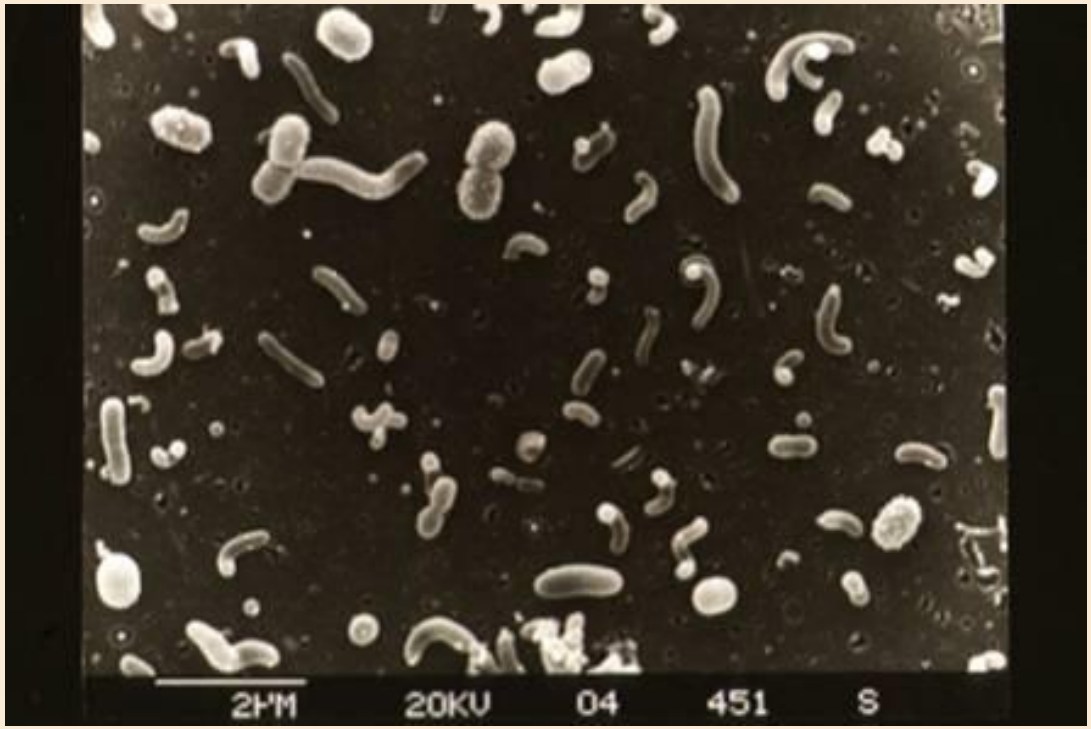
Archaea

Many are extremophiles…



Archaea

Even so, archaea are not restricted to extreme habitats



Archaea account for 20-30% of marine microbial cells

Comparison of Three Cellular Domains

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TABLE 4.5 Comparison of Three Cellular Domains

Characteristic	Bacteria	Archaea	Eukarya
Cell type	Prokaryotic	Prokaryotic	Eukaryotic
Chromosomes	Single, or few, circular	Single, circular	Several, linear
Types of ribosomes	70S	70S but structure is similar to 80S	80S
Contains unique ribosomal RNA signature sequences	+	+	+
Number of sequences shared with Eukarya	One	Three	(All)
Protein synthesis similar to Eukarya	−	+	
Presence of peptidoglycan in cell wall	+	−	−
Cell membrane lipids	Fatty acids with ester linkages	Long-chain, branched hydrocarbons with ether linkages	Fatty acids with ester linkages
Sterols in membrane	− (some exceptions)	−	+

So…..

Microbial functional diversity is a constituent of biodiversity that includes wide range of metabolic activities that can influence numerous aspects of ecosystem functioning like ecosystem stability, nutrient availability, ecosystem dynamics, etc.

Microbial diversity is essential

1. for human survival and economic security in the future, as it provides a diverse range of resources that humans can use to their advantage.
2. Understanding microbial diversity can aid in the mitigation of new threats such as diseases, pathogens, and viruses.